

The mixed glass former effect in twin-roller quenched lithium borophosphate glasses

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ABSTRACT

A whole series of lithium borophosphate glasses of composition $45 \text{ Li}_2\text{O}-55 [x \text{ B}_2\text{O}_3-(1-x) \text{ P}_2\text{O}_5]$, $0 \leq x \leq 1$ was prepared for the first time, thanks to the use of the twin roller quenching technique. As their more popular sodium counterparts, these glasses exhibit a mixed glass former effect with non linear increase in both glass transition temperature and conductivity when phosphorous is replaced by boron. A fairly good conductivity ($\sim 2 \times 10^{-7} \Omega^{-1} \cdot \text{cm}^{-1}$ at room temperature) coupled to a high T_g ($\sim 450^\circ \text{C}$) makes these glasses interesting as solid electrolytes for the development of thin-film batteries. A structural investigation was carried out both by Raman spectroscopy and for the first time, by ^{11}B NMR at high field, 18.8 T. Raman spectra showed the signature of many types of vibrations indicating the presence of a large variety of entities in the glasses. ^{11}B NMR at 18.8 T helped in separating the resonances of three-coordinated and four-coordinated boron species and counting them by direct integration. On the whole it was shown that while the T_g evolution relied upon the presence of mixed borophosphate entities, the conductivity was directly linked to the presence of BO_4 entities alone irrespective of their specific environment.

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1. Introduction

Lithium-ion based glassy electrolytes have been extensively studied as they are potential candidates for the development of all solid state batteries [1–4]. Indeed, the use of glassy solid electrolytes could avoid some drawbacks encountered with the use of liquid electrolytes such as corrosion of the electrodes and makes the manufacturing of the devices easier by comparison with their crystalline counterparts [5–8]. The miniaturization of the batteries can then also be considered [9]. Nevertheless, in order to be used in commercialized devices, solid electrolytes must have high ionic conductivity and above all good chemical, thermal and electrochemical stability [10]. Alkali/silver borophosphate glasses have been widely studied during the last decades. These systems are well known to present the so-called “mixed-glass former effect”, i.e. an enhancement of conductivity when a former cation (B or P) is replaced by the other one (P or B), the total amount of modifier being constant. Such enhancement in the ionic conductivity usually goes along with an increase in the glass transition temperature and also in the thermal stability [11–14]. Many papers also report studies about the local environment in the glass using FTIR, Raman and NMR techniques [11–20]. It is repeatedly suggested that the existence of BPO_4 units in the ternary glasses might explain the enhancement in conductivity. However, the literature evidence for BPO_4 -like environments in these glasses,

particularly with respect to the short range order of the P-atoms is not convincing [18].

Even though lithium borophosphate glasses would be the most interesting materials in regards to applications, they are the least studied. It can be understood by the fact that these glasses show a high tendency to crystallize, and therefore a limited glass forming region [12,13,15]. In the present work, the system $45 \text{ Li}_2\text{O}-55 [x \text{ B}_2\text{O}_3-(1-x) \text{ P}_2\text{O}_5]$, $0 \leq x \leq 1$, was studied. The innovation of this work was the use of the twin roller quenching technique as an attempt to extend the vitreous domain by preventing crystallization during cooling. The number of potential charge carriers remained constant in order to study the influence of the mixed glass former effect on the physical properties, especially the ionic conductivity. Raman experiments and, for the first time, an ^{11}B MAS NMR investigation at high field (18.8 T), were carried out to study the local structure of the glass.

2. Experimental

Glasses of composition $45 \text{ Li}_2\text{O}-55 [x \text{ B}_2\text{O}_3-(1-x) \text{ P}_2\text{O}_5]$ were first prepared by the conventional melt quenching procedure ($\sim 10^2 \text{ K.s}^{-1}$). Reagent grade raw materials Li_2CO_3 (Aldrich, 99%), $\text{NH}_4\text{H}_2\text{PO}_4$ (Aldrich, 99.999%) and H_3BO_3 (Aldrich, 99.5%) were mixed in stoichiometric amounts. The batches were heated at 600°C in a platinum crucible to remove all the volatile products. They were further melted at 1100°C . Homogeneous melt was poured and quenched on a brass plate.

Samples were then prepared by the twin roller quenching method in an argon filled glove box. This technique allows reaching a

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quenching speed of about 10^6 K.s^{-1} [21]. In this case, the already synthesized compounds were melted again in an induction-heated platinum crucible. The melt dropped between two rollers, and flakes of about $60 \mu\text{m}$ in thickness were then obtained.

X-ray Diffraction (XRD) analysis of the compounds was carried out with a PANalytical X-PERT diffractometer using monochromatic $\text{Cu K}\alpha$ radiation ($1.5418 \text{ \AA}$) within the range $10^\circ < 2\theta < 60^\circ$.

Fresh fractures of glasses were investigated by Field Emission Scanning Electron Microscopy (FE-SEM) with a Hitachi S-4800 microscope. The glasses were fractured in a glove box and rapidly inserted into the microscope in order to limit the contact with air.

Glass transition temperatures T_g were measured by Differential Scanning Calorimetry (DSC) with a Mettler Toledo DSC-1 instrument at a constant heating rate of 10 K.min^{-1} using powdered samples placed in an alumina capsule. Capsules were prepared in a glove box. T_g values were taken at the onset of the endothermic effect on the DSC pattern.

Electrical conductivity measurements were performed by Complex Impedance Spectroscopy (CIS) using a Solartron SI1260 impedance analyzer, in the frequency range $10 \text{ Hz} - 10^7 \text{ Hz}$ at temperatures comprised between 30°C and 170°C , with an applied voltage of 0.5 V . The flakes obtained after twin roller quenching were sputtered while covered with a mask in order to obtain platine electrodes of 5 mm in diameter.

Raman spectra were recorded on a HORIBA Jobin Yvon LabRAM ARAMIS IR2 spectrometer ($\lambda = 473 \text{ nm}$) and the data were acquired with an integration time of 30 s .

The presence of crystalline BPO_4 in some re-crystallized glasses was investigated by 1D ^{31}P MAS-NMR performed at 9.4 T ($\nu_0(^{31}\text{P}) = 162 \text{ MHz}$) using a 3.2 mm HXY probe operating at a spinning frequency of 20 kHz . A single pulse sequence with a pulse length of $4 \mu\text{s}$ and a rf field of 62.5 kHz was used, 32 transients separated by a rd of 4000 s were recorded.

The structure of the glasses was investigated by 1D Magic Angle Spinning Nuclear Magnetic Resonance (MAS-NMR) performed on Bruker Avance II spectrometers. A very high field of 18.8 T ($\nu_0(^{11}\text{B}) = 256.81 \text{ MHz}$) was used in order to improve the resolution of the ^{11}B spectra since the ^{11}B is a quadrupolar nucleus. The experiments were carried out with a 3.2-mm HXY probe operating at a spinning frequency of 20 kHz . A single pulse sequence with a pulse length of $1 \mu\text{s}$ and an rf amplitude corresponding to an ^{11}B nutation frequency of 40 kHz was used and 1024 transients separated by a recycle delay (rd) of 3 s were recorded. The baseline was corrected by subtracting the background signal coming from the Boron Nitride stator of the probe.

3. Results and discussion

The $\text{Li}_2\text{O}-\text{B}_2\text{O}_3-\text{P}_2\text{O}_5$ phase diagram is shown in Fig. 1 along with the studied compositions $45 \text{ Li}_2\text{O}-55 [x \text{ B}_2\text{O}_3-(1-x) \text{ P}_2\text{O}_5]$, $0 \leq x \leq 1$, represented as circle symbols and the crystalline phases labeled with a star.

3.1. Extension of vitreous domain

The amorphous state of the prepared samples was checked by XRD. Several significant XRD patterns are reported in Fig. 2. When the samples were prepared by the conventional quenching route, the vitreous domain extended from $0 \leq x \leq 0.55$ (filled circles in Fig. 1). As shown in Fig. 2, the patterns of the compositions $x = 0.64$ and 0.73 indeed show the presence of Bragg peaks corresponding to the crystalline phase Li_3PO_4 . When the samples were prepared by the twin roller quenching technique the vitreous domain was extended up to $x = 1$ (empty circles in Fig. 1) as shown in Fig. 2.

A FE-SEM investigation confirmed these results. Fig. 3 shows the FE-SEM micrographs for $x = 0.64$ and $x = 0.73$ samples prepared by conventional quenching. The micrograph of the conventionally

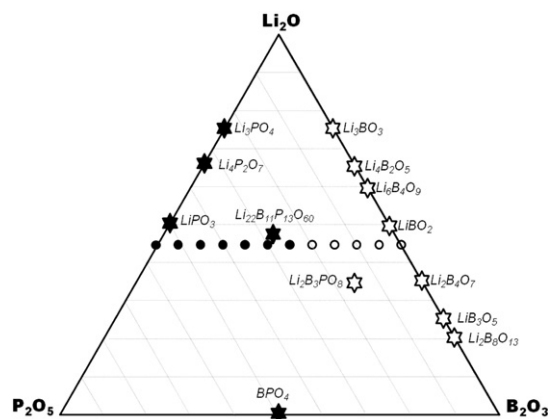


Fig. 1. $\text{Li}_2\text{O}-\text{B}_2\text{O}_3-\text{P}_2\text{O}_5$ ternary diagram showing the studied compositions: conventionally quenched samples (●) and twin-roller quenched samples (○). The crystalline phases that appeared when annealing the glasses at 100°C above T_g for 24 h (★) and other existing crystalline phases (☆) are also reported.

quenched sample $x = 0.64$ showed a lot of “aggregates” of about 90 nm in size which is in agreement with the fact that the sample is partially crystallized. The micrograph of the conventionally quenched sample $x = 0.73$ is rather the signature of a sample where the dominant phase is a crystalline one with the presence of needle-shape crystallites.

Fig. 4 exhibits micrographs of twin roller quenched glasses. The twin roller quenched samples appear to be perfectly homogenous and there is no evidence of any crystallization or phase separation for all the compositions (all the micrographs are not shown herein).

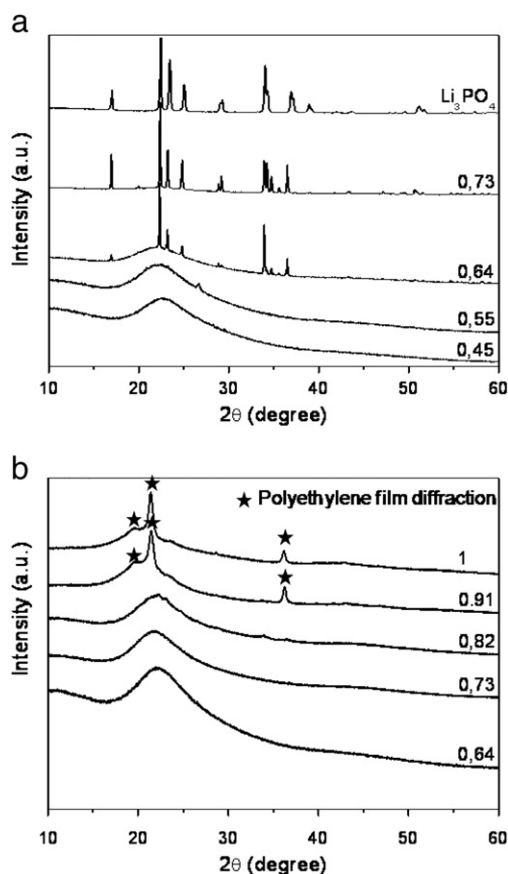


Fig. 2. X-Ray diffraction patterns of the glasses $45 \text{ Li}_2\text{O}-55 [x \text{ B}_2\text{O}_3-(1-x) \text{ P}_2\text{O}_5]$ for the conventional quenched samples and crystalline Li_3PO_4 (a) and the twin roller quenched samples (b). (A polyethylene film was used to protect air sensitive samples).

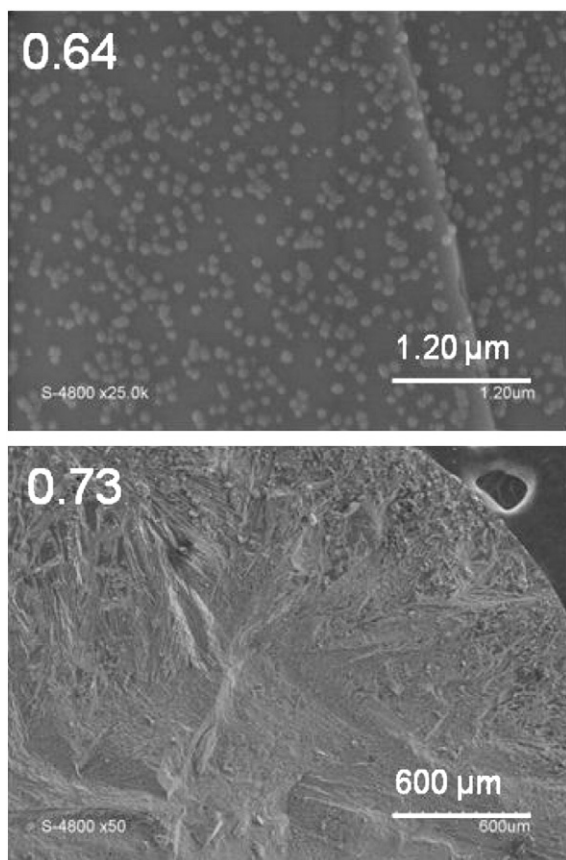


Fig. 3. FE-SEM micrographs of the glasses 45 Li₂O–55 [x B₂O₃–(1–x) P₂O₅] for the x = 0.64 and x = 0.73 samples prepared by conventional quenching.

In order to clarify the crystallization tendency of these borophosphate compounds, the samples were annealed at 100 °C above the glass transition temperature for 24 h. Fig. 5 presents the X-Ray Diffraction patterns of the annealed samples. The crystalline phases are indicated by filled stars in the phase diagram given in Fig. 1. The crystallized samples can be divided in three series. At low boron content, for x = 0 and 0.09, only phosphate phases crystallized, i.e. LiPO₃ and LiPO₃ + Li₆P₆O₁₈ respectively. Then for x = 0.18 to x = 0.45, borophosphate phases started appearing. Essentially three crystalline phases BPO₄, Li₄P₂O₇ and Li₂₂B₁₁P₁₃O₆₀ could be found in this region. ³¹P NMR spectra for x = 0.36 and 0.45 were recorded and are shown in Fig. 6 along with those of the crystalline phases BPO₄ and Li₄P₂O₇. BPO₄ is only present in sample x = 0.36. No clear signature for the phase Li₂₂B₁₁P₁₃O₆₀ is observed in the NMR spectra, unless it corresponds to the broad signal ranging from 0 ppm to –25 ppm which would indicate a very disordered local environment for P in this phase. Finally at high boron content, phosphate phases alone crystallized again while the borophosphate phases did not crystallize anymore. Only Li₄P₂O₇ and Li₃PO₄ were present for x = 0.55 while Li₃PO₄ was the only phase present for the B-rich samples. This last result agrees with the fact that the same phase crystallized in the samples prepared by conventional air quenching beyond x = 0.55.

It is interesting to note that the spontaneous crystallization upon air-quenching occurred for compositions in the B-rich region where neither borate nor borophosphate crystalline phase appeared after a 24 h-annealing above T_g.

On the whole, when a conventional air quenching procedure was used, the present results are in the line of those reported by Muñoz et al. [13] for similar compounds, i.e. a vitreous domain in the range 0 ≤ x ≤ 0.55 for 45Li₂O–55[xB₂O₃–(1–x)P₂O₅] and 0 ≤ x ≤ 0.40 for 50 Li₂O–50 [xB₂O₃–(1–x)P₂O₅] respectively. The use of twin roller

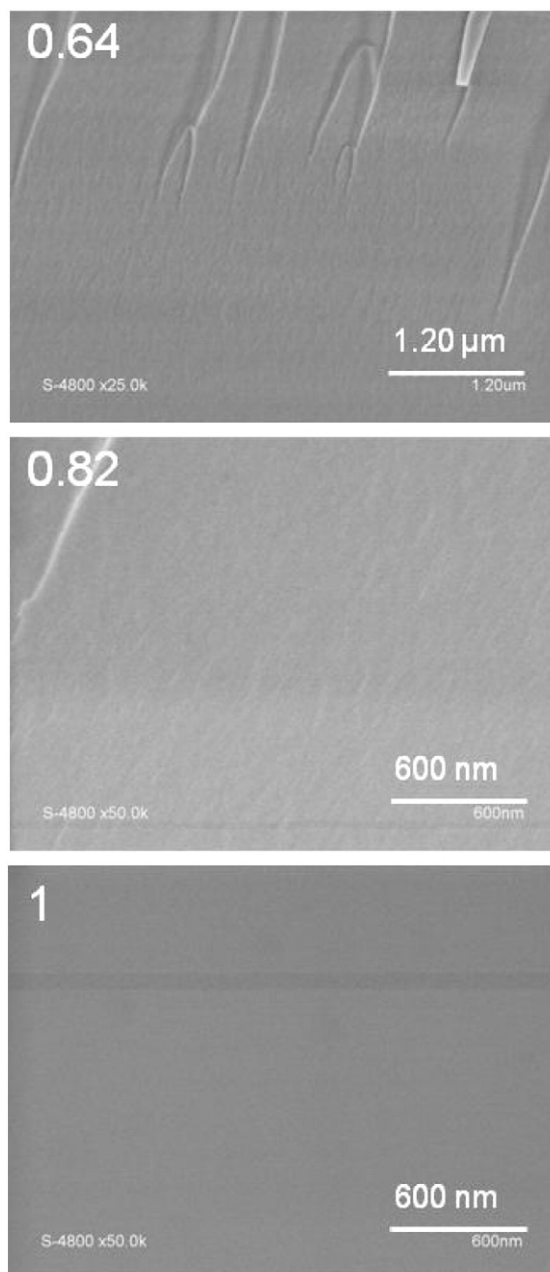


Fig. 4. FE-SEM micrographs of the glasses 45 Li₂O–55 [x B₂O₃–(1–x) P₂O₅] for the x = 0.64, x = 0.82 and x = 1 samples prepared by twin-roller quenching.

quenching allowed to extend the vitreous domain all along the line i.e. up to x = 1.

This result allowed carrying out the thermal, electrical and structural characterization of a complete set of high Li⁺ content borophosphate glasses for the first time.

3.2. Thermal and electrical characterization

Table 1 gathers the glass transition temperatures (T_g), the electrical conductivities (σ) and the activation energies (E_a) of the borophosphate glasses for samples prepared by conventional and twin roller quenching. For a given composition, when a glass could be produced, data are similar whatever the quenching procedure used. Whenever a comparison can be done, the present data are close to the values reported by Muñoz et al. [13] and by Money et al. [12] for similar compositions.

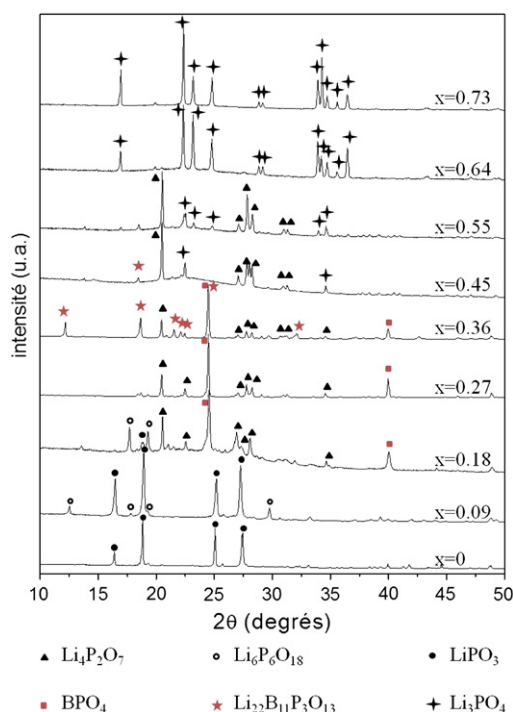


Fig. 5. X-Ray Diffraction patterns of samples 45 Li₂O–55 [x B₂O₃–(1–x) P₂O₅] from x=0 to x=0.73 annealed for 24 h at 100 °C above T_g.

Fig. 7 shows the variation of T_g of the 45 Li₂O–55 [x B₂O₃–(1–x) P₂O₅] glasses as a function of the boron content while Fig. 8 depicts the variation of the logarithm of the electrical conductivities (σ) at room temperature and the activation energies (E_a) for both quenching routes.

The replacement of P₂O₅ by B₂O₃ in the glassy matrix led to an increase in the T_g values. In the same time, an important increase in conductivity for an increasing amount of boron from x=0 to x=0.45 was observed, then the increase slowed down for samples x=0.45 to x=0.82. Finally from x=0.91 to x=1, the conductivity slightly decreased. On the whole, the evolution is similar to that reported by Zielniok et al. [20] on Na⁺ borophosphate glasses with a shift of the curves due to a change in both the nature (Li⁺/Na⁺) and amount (45/40) in cation. These results are the signature of a mixed-former effect. In spite of a constant amount of potential charge carriers Li⁺ a non-linear enhancement in log σ was observed when phosphorus was substituted by boron. As usually found in systems showing the mixed former effect, the increase in conductivity went

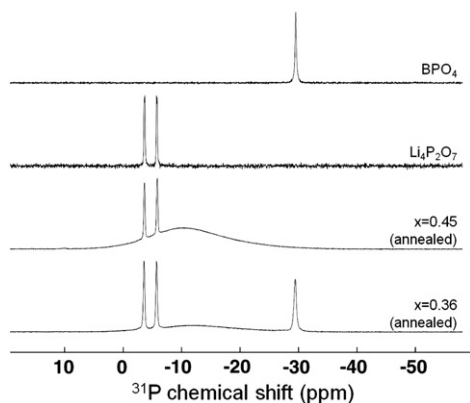


Fig. 6. ³¹P NMR spectra of samples 45 Li₂O–55 [x B₂O₃–(1–x) P₂O₅], x = 0.36 and 0.45, annealed for 24 h at 100 °C above T_g. ³¹P NMR spectra of crystalline BPO₄ and Li₄P₂O₇ are shown for comparison.

Table 1

Glass transition temperatures, conductivity values at room temperature and activation energies for both conventionally quenched and twin roller quenched compounds with compositions 45 Li₂O–55 [x B₂O₃–(1–x) P₂O₅].

x	Conventionally quenched glasses			Twin roller quenched glasses		
	T _g (°C)	σ (Ω ⁻¹ .cm ⁻¹) at RT	E _a (eV)	T _g (°C)	σ (Ω ⁻¹ .cm ⁻¹) at RT	E _a (eV)
0	300	3.0 × 10 ⁻¹⁰	0.80	300	3.0 × 10 ⁻¹⁰	0.85
0.09	329	1.9 × 10 ⁻⁹	0.75	341	2.8 × 10 ⁻⁹	0.75
0.18	353	1.0 × 10 ⁻⁸	0.68	362	9.3 × 10 ⁻⁹	0.71
0.27	382	3.2 × 10 ⁻⁸	0.65	385	3.5 × 10 ⁻⁸	0.68
0.36	407	5.8 × 10 ⁻⁸	0.62	410	6.8 × 10 ⁻⁸	0.65
0.45	417	1.2 × 10 ⁻⁷	0.59	418	1.3 × 10 ⁻⁷	0.61
0.55	419	9.3 × 10 ⁻⁸	0.60	426	1.5 × 10 ⁻⁷	0.63
0.64	–	–	–	434	2.0 × 10 ⁻⁷	0.61
0.73	–	–	–	450	2.4 × 10 ⁻⁷	0.60
0.82	–	–	–	453	2.5 × 10 ⁻⁷	0.58
0.91	–	–	–	461	1.9 × 10 ⁻⁷	0.60
1	–	–	–	435	5.9 × 10 ⁻⁸	0.63

along with an increase in the vitreous transition temperature. A decrease in T_g is usually observed in binary glasses when the amount in modifier increases and it is attributed to a depolymerization of the network. In the ternary system under study, the DSC data indicate that the richer in boron the glass is, the higher the T_g is and therefore the more the network is polymerized. The increase in conductivity can then hardly be assigned to a depolymerization phenomenon which would make the mobility of cations easier.

3.3. Structural characterization

Fig. 9 gathers the Raman spectra of 45 Li₂O–55 [x B₂O₃–(1–x) P₂O₅] glasses. Similar Raman spectra for 50 Li₂O–50 [x B₂O₃–(1–x) P₂O₅] glasses in the domain where air-quenched glasses were achievable, were reported by Muñoz et al. [13]. The spectra are also similar to earlier data obtained for air-quenched alkali/silver borophosphate glasses [10,12–14,19]. The present analysis is based upon these previous reports.

The glass corresponding to x=0 has a composition close to the phase LiPO₃. Thus, the two contributions observed at 1180 cm⁻¹ and 690 cm⁻¹ can be attributed to the symmetric O–P–O and P–O–P stretching vibrations in PO₄ tetrahedra and (PO₃)_n⁻ chains respectively as it was observed by Money et al. [12]. As already proposed by Zielniok et al. [20] for sodium borophosphate glasses, the broad band centered at about 1290 cm⁻¹ is attributed to P=O stretching vibrations. The intensity of this line decreases continuously when x increases. From x=0 to x=0.18, the O–P–O contribution at

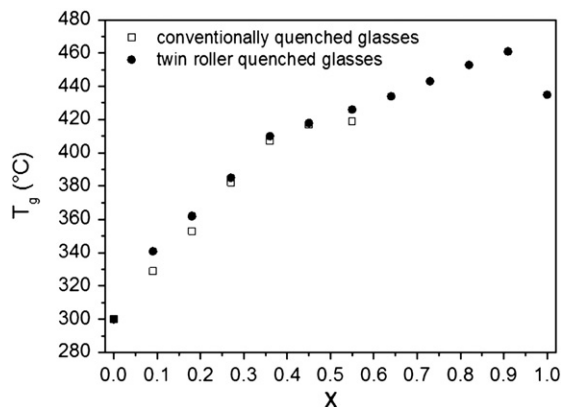


Fig. 7. Glass transition temperatures of the glasses 45 Li₂O–55 [x B₂O₃–(1–x) P₂O₅] as a function of x.

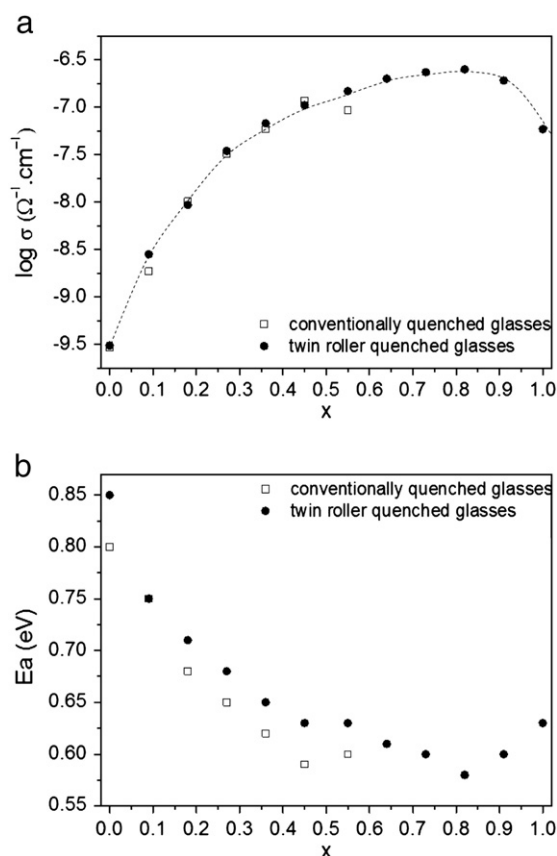


Fig. 8. Logarithm of conductivity at room temperature (a) and activation energies (b) of samples 45 Li₂O–55 [x B₂O₃–(1–x) P₂O₅] as a function of x for both quenching routes.

1180 cm^{−1} keeps broadening and simultaneously shifts to lower wave numbers (~1150 cm^{−1}).

From $x = 0.18$, one can see the emergence of two other contributions at 650 cm^{−1} and 1100 cm^{−1}. They account for the creation of B–O–P bonds in the glass network [12,17]. The line at 1100 cm^{−1} is predominantly intense in the intermediate composition ($x = 0.36$). At the opposite of the line at 650 cm^{−1} it is still present for $x > 0.55$ and up to $x = 1$ indicating that it also comprises vibrations due to boron entities alone. For $x = 0.27$, this band and the vibrations of the O–P–O bonds overlap. For higher values of x , the O–P–O contribution vanishes and so, the P–O–P band at 690 cm^{−1}. For $0.36 \leq x \leq 0.82$, the line at 1100 cm^{−1} overlaps with a band at 1000 cm^{−1} which is attributed to (P₂O₇)^{4−} and (B–O–P)[−] entities [19].

Moreover a formation of B₄O₇^{2−} units is suggested by the appearance of a band centered at about 770 cm^{−1} [12] clearly observable from $x = 0.27$ up to $x = 1$. The contribution centered at 950 cm^{−1} which is clearly visible at high boron content, is assigned to the formation of four-coordinated boron BO₄[−] in Li⁺BO₄[−] units [12]. Finally one can see the emergence of the B–O stretching vibrations of the B–O bonds in the BO₃ units at 1430 cm^{−1} from $x = 0.82$ to $x = 1$ [9].

It is difficult to go further in the interpretation of these spectra. Clearly, the structure of the glasses is very complex with many types of species including mixed borophosphate units at least in the intermediate region. In order to identify the different borate species experiments by ¹¹B NMR spectroscopy on an 18.8 T spectrometer were carried out.

¹¹B NMR investigation had already been used to identify the boron entities present in the borophosphate glasses [13,20]. However to date, all the data were recorded at fields where a complete separation between three-coordinated and four-coordinated boron could not be

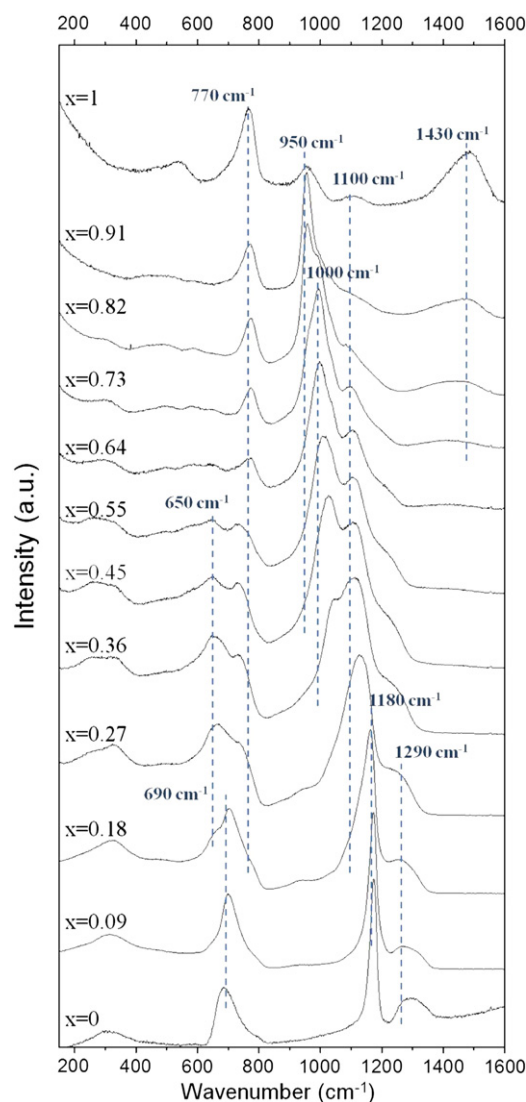


Fig. 9. Raman spectra of the 45 Li₂O–55 [x B₂O₃–(1–x) P₂O₅] glasses for $x = 0$ to 1.

achieved. In the present investigation, the high magnetic field used for the experiments, i.e. 18.8 T, allowed for this separation. The ¹¹B spectra reported in Fig. 10 show the presence of three-coordinated (centered between 20 and 8 ppm) and four-coordinated (centered between 4 and −7 ppm) boron signals. At low boron concentrations ($x = 0.09$ – 0.27), the spectra only exhibit signals characteristic of

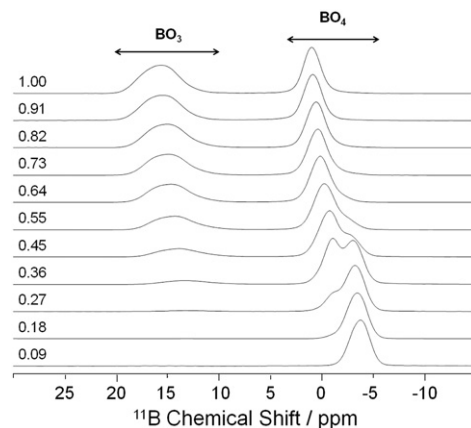


Fig. 10. ¹¹B NMR spectra of the 45 Li₂O–55 [x B₂O₃–(1–x) P₂O₅] glasses for $x = 0$ to 1.

tetra-coordinated boron. Two distinct BO_4 signals can be observed: a first resonance at -4 ppm gradually replaced by a second one centered at -1 ppm when the content of B_2O_3 increases in the formulation. It is also worthy to note that the chemical shift of the second signal slightly changes to reach 0 ppm in case of the pure borate composition ($x = 1$), suggesting a progressive evolution of the chemical environment of this particular species, as previously reported by Zielniok et al. [20]. The authors stated that the chemical shift evolution is related to a progressive replacement of $\text{B}-\text{O}-\text{P}$ by $\text{B}-\text{O}-\text{B}$ bonds. If this progressive replacement can explain the linear evolution of the chemical shift, it can hardly support the existence of two distinct BO_4 signals. Additional advanced NMR experiments are required to clarify this point [22]. The broad resonance with a quadrupolar line-shape characteristic of BO_3 appears on the $x = 0.27$ sample spectrum and increases with the B content. These data are in agreement with both those by Muñoz et al. [13] and Zielniok et al. [20].

The well-resolved spectra allow for an easy and efficient BO_3/BO_4 quantification by simple signal integration. The borate species, BO_3 and BO_4 , were quantified using the number of boron derived from the formulation $45\text{Li}_2\text{O}-55[x\text{B}_2\text{O}_3-(1-x)\text{P}_2\text{O}_5]$. These numbers are shown in Fig. 11. The overall BO_4 evolution starts with an important increase for samples with moderate amounts of boron ($0 \leq x \leq 0.36$) but only experiences a limited increase for glasses containing high boron contents ($0.36 \leq x \leq 0.81$). The BO_3 signal starts being significantly observed for $x = 0.27$. This signal then experiences an exponential increase with the boron content to become the main boron contribution for $x > 0.81$ samples.

3.4. The mixed glass former effect

As it was previously shown by Zielniok et al. [23], there is a close relationship between the electrical conductivity behavior and the number of four-coordinated boron atoms. Figs. 8 and 11 show that the conductivity curve is very similar to the BO_4 curve, as long as BO_3 entities are not too many. When the BO_3/BO_4 ratio is larger than 1, the conductivity starts decreasing. A possible interpretation of the previous finding is the following: the conductivity would be mainly controlled by Li^+ ions involved in the charge compensation of BO_4 . Note that these ions are more mobile than Li^+ linked to non-bridging oxygen. The initial increase in conductivity with increasing x would then come from the gradual replacement of anionic metaphosphate units (which serve as stronger Coulombic traps) by the BO_4 units, whose charge is more delocalized. At high boron content, the BO_3 entities start dominating. They would then break the conduction pathways formed by the BO_4 entities.

On the whole these data show that the conductivity is directly linked to the number of four-coordinated boron species irrespective of their specific environment. The presence of mixed borophosphate

species has probably no direct influence on the conductivity but it helps in the reticulation of the network as observed by the increase in T_g and probably in the stabilization of the glasses. Remember that the spontaneous crystallization of the compounds upon air quenching occurs when the phosphate phases starts dominating the crystallization.

4. Conclusion

The twin roller quenching technique allowed for the whole series of lithium borophosphate glasses with composition $45 \text{Li}_2\text{O}-55 [x \text{B}_2\text{O}_3-(1-x) \text{P}_2\text{O}_5]$, $0 \leq x \leq 1$ to be synthesized for the first time. These glasses exhibited a mixed glass former effect with the vitreous transition temperature and the conductivity showing non linear variations when replacing phosphorous by boron in the glassy network. Structural investigation included Raman and for the first time for borophosphate glasses, ^{11}B NMR at a high field, 18.8 T . The Raman spectra exhibited the signature of many types of vibrations indicating the presence of a large variety of entities in the glasses. ^{11}B NMR at 18.8 T helped in separating the resonances of three-coordinated and four-coordinated boron species and counting them by direct integration for the first time. On the whole it was shown that while the T_g evolution relied upon the presence of mixed borophosphate entities, the conductivity was directly linked to the presence of BO_4 entities alone irrespective of their specific environment.

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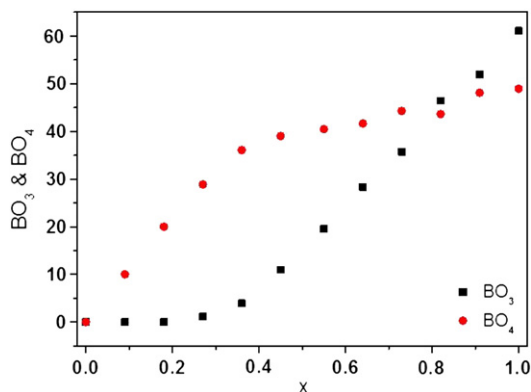


Fig. 11. Number of BO_3 and BO_4 entities for the $45 \text{Li}_2\text{O}-55 [x \text{B}_2\text{O}_3-(1-x) \text{P}_2\text{O}_5]$ glasses, $x = 0$ to 1 .